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GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Logic Gates

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

40	26	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	26 questions
Marks 2	10 questions
Marks 5	3 questions
Marks 8	1 question
Answer key	at the end

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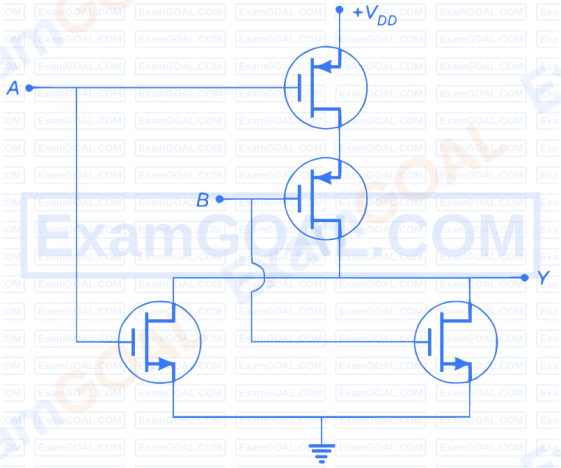
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 26 questions

Q1. GATE ECE 2026

In the circuit shown in the Figure, A and B are logic inputs and Y is the logic output. Which of the following logic operations is realized by the circuit?

Open solution



(A) NOR (B) OR (C) XOR (D) AND

Q2. GATE ECE 2025

A 3-input majority logic gate has inputs X, Y and Z. The output F of the gate is logic ' 1 ' if two or more of the inputs are logic ' 1 '. The output F is logic ' 0 ' if two or more of the inputs are logic '0'. Which one of the following options is a Boolean expression of the output F ?

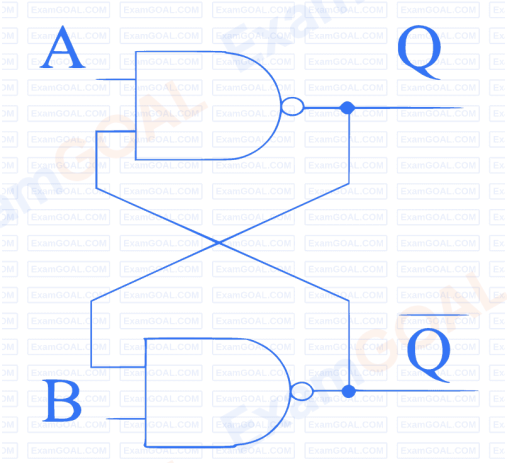
Open solution

(A) $XY + YZ + ZX$ (B) $X \oplus Y \oplus Z$ (C) $X + Y + Z$ (D) XYZ

Q3. GATE ECE 2023

For the circuit shown below, the propagation delay of each NAND gate is 1 ns. The critical path delay, in ns, is _____ (rounded off to the nearest integer).

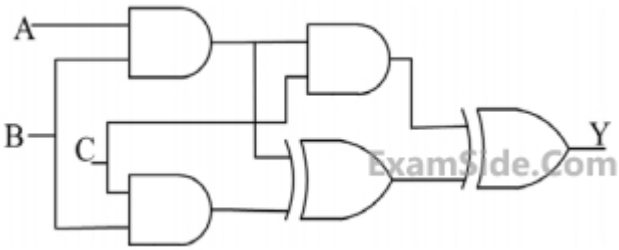
Open solution



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Q4. GATE ECE 2016 Set 1

The output of the combinational circuit given below is



Open solution

(A) A+B+C (B) A(B+C) (C) B(C+A) (D) C(A+B)

Q5. GATE ECE 2016 Set 3

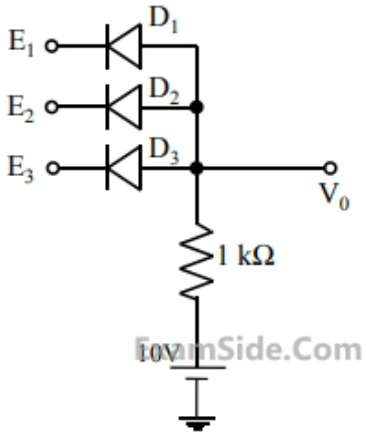
The minimum number of 2-input NAND gates required to implement a 2-input XOR gate is

(A) 4 (B) 5 (C) 6 (D) 7

Open solution

Q6. GATE ECE 2015 Set 3

In the circuit shown, diodes D_1 ,D_2 and D_3 are ideal, and the inputs E_1 , E_2 and E_3 are “0 V” for logic ‘0’ and “10 V” for logic ‘1’. What logic gate does the circuit represent?



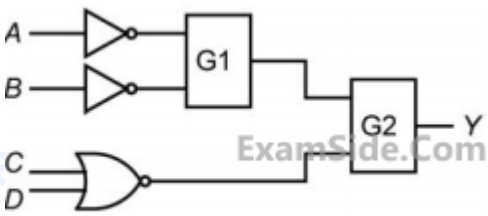
Open solution

(A) 3-input OR gate (B) 3-input NOR gate
(C) 3-input AND gate (D) 3-input XOR gate

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Q7. GATE ECE 2015 Set 2

In the figure shown, the output Y is required to be Y = AB + $\overline{C}\,\overline{D}$. The gates G1 and G2 must be, respectively,

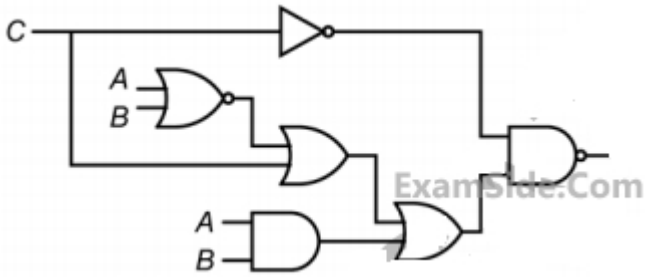


Open solution

(A) NOR, OR (B) OR, NAND (C) NAND, OR (D) AND, NAND

Q8. GATE ECE 2014 Set 4

In the circuit shown in the figure, if C = 0, the expression for Y is



Open solution

(A) Y= $A\overline{B} + \overline{A}B$ (B) Y=A+B

(C) Y= $\overline{A}\,\overline{B}$ (D) Y=A B

Q9. GATE ECE 2013

A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles.

Open solution

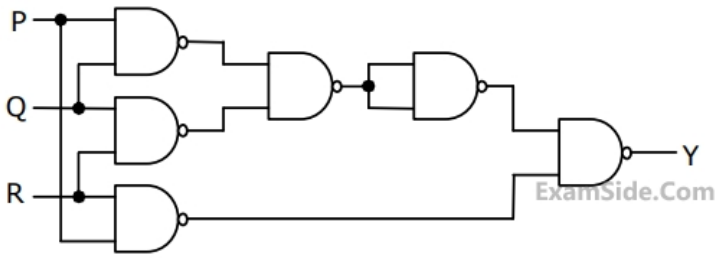
(A) an AND gate (B) an OR gate (C) an XOR gate (D) a NAND gate

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Q10. GATE ECE 2011

Open solution

The output Y in the circuit below is always '1' when



(A) two or more of the inputs P, Q, R are '0'

(B) two or more of the inputs P, Q, R are '1

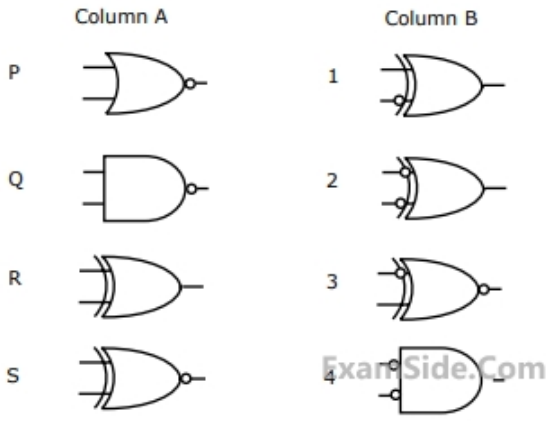
(C) any odd number of the inputs P, Q, R is '0'

(D) any odd number of the inputs P, Q, R is '1'

Q11. GATE ECE 2010

Open solution

Match the logic gates in column A with their equivalents in column B.



(A) P-2, Q-4, R-1, S-3

(B) P-4, Q-2, R-1, S-3

(C) P-2, Q-4, R-3, S-1

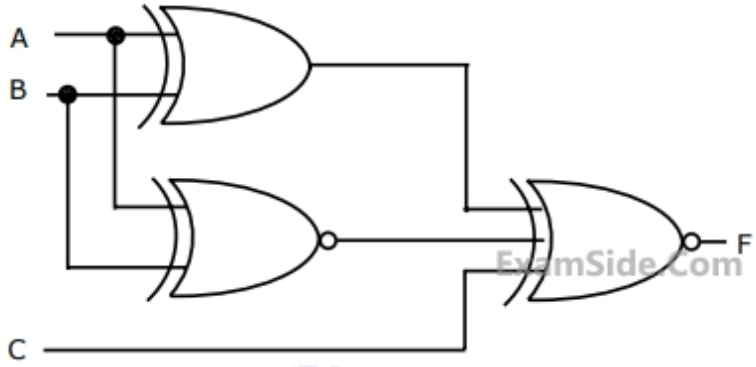
(D) P-4, Q-2, R-3, S-1

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Q12. GATE ECE 2010

Open solution

For the output F to be 1 in the logic circuit shown, the input combination should be



(A) A = 1, B= 1. C = 0

(B) A = 1, B= 0,C = 0

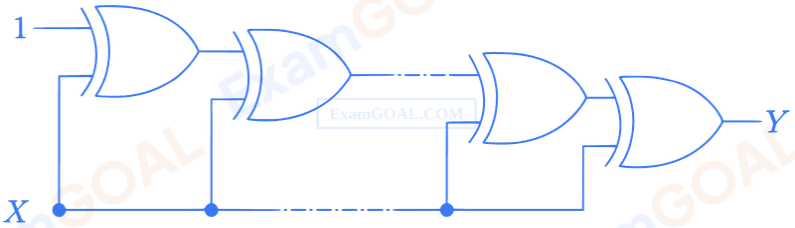
(C) A = 0, B= 1. C = 0

(D) A = 0, B= 0, C = 1

Q13. GATE ECE 2002

Open solution

If the input to the digital circuit (in the figure) consisting of a cascade of 20 XOR-gates is X then the output Y is equal to



(A) 0

(B) 1

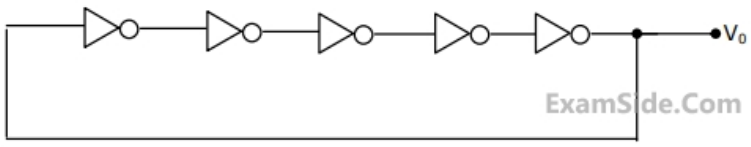
(C) $\overline{X}$

(D) X

Q14. GATE ECE 2001

Open solution

For the ring oscillator shown in the figure, the propagation delay of each inverter is 100 pico sec. What is the fundamental frequency of the oscillator output?



(A) 10 MHz

(B) 100 MHz

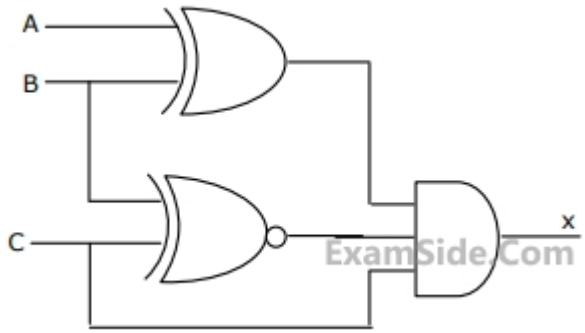
(C) 1 GHz

(D) 2 GHz

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Q15. GATE ECE 2000

For the logic circuit shown in Figure, the required input condition (A, B, C) to make the output (X)=1.

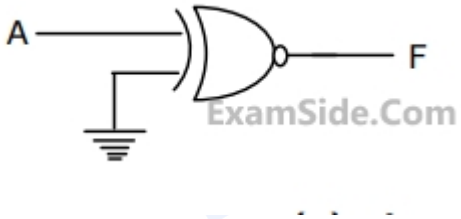


Open solution

(A) 1, 0, 1 (B) 0, 0, 1 (C) 1, 1, 1 (D) 0, 1, 1

Q16. GATE ECE 1997

The output of the logic gate in figure is



Open solution

(A) 0 (B) 1 (C) $\overline{A}$ (D) A

Q17. GATE ECE 1995

The minimum number of NAND gates required to implement the Boolean function $A + A\overline{B} + A\overline{B}C$ is equal to

(A) Zero (B) 1 (C) 4 (D) 7

Open solution

Q18. GATE ECE 1994

A ring oscillator consisting of 5 inverters is running at a frequency of 1.0 MHz. The propagation delay per gate is _____ ns.

Open solution

Q19. GATE ECE 1993

Boolean expression for the output of XNOR (equivalence) logic gate with inputs A and B is

(A) $A\overline{B} + \overline{A}B$ (B) $\overline{A}\,\overline{B} + AB$

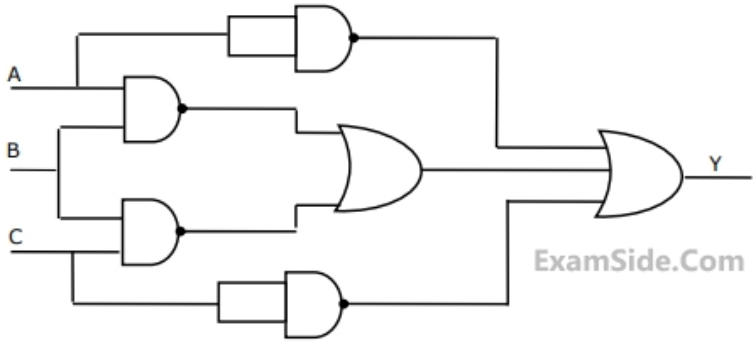
(C) $(\overline{A} + B)(A + \overline{B})$ (D) $(\overline{A} + \overline{B})(A + B)$

Open solution

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Q20. GATE ECE 1993

For the logic circuit shown in Figure, the output is equal to



Open solution

(A) $\overline{A}.\overline{B}.\overline{C}$

(B) $\overline{A}+\overline{B}+\overline{C}$

(C) $\overline{AB}+\overline{BC}+\overline{A}+\overline{C}$

(D) $\overline{AB}+\overline{BC}$

Q21. GATE ECE 1989

Indicate which of the following logic gates can be used to realize all possible combinational Logic functions:

Open solution

(A) OR gates only.

(B) NAND gates only.

(C) EX-OR gates only.

(D) NOR gates only.

Q22. GATE ECE 1988

Minimum number of 2-input NAND gates required to implement the function, f=$(\overline{X}+\overline{Y})$(Z+W) is

Open solution

(A) 3

(B) 4

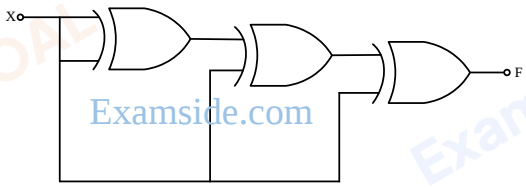
(C) 5

(D) 6

Q23. GATE ECE 1988

For the circuit shown below the output F is given by

Open solution



(A) F=1

(B) F=0

(C) F=X

(D) F= $\overline{X}$

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Q24. GATE ECE 1988

The Boolean function A+BC is a reduced form of

(A) AB+BC
(B) (A+B).(A+C)
(C) $\overline{A}$B+A$\overline{B}$ C
(D) (A+C).B

Open solution

Q25. GATE ECE 1988

The minimum number of 2-input NAND gates required to implement the Boolean function Z=A$\overline{B}$C, assuming that A, B and C are available, is

(A) two
(B) three
(C) five
(D) six

Open solution

Q26. GATE ECE 1988

For the identity AB+$\overline{A}$ C + BC= AB + $\overline{A}$ C, The dual form is

(A) (a+b)($\overline{A}$ + c)(b+c)=(a+b)($\overline{A}$+C)
(B) $(\overline{A}+\overline{B})(A+\overline{C})(\overline{B}+\overline{C})=(\overline{A}+\overline{B})(A+\overline{C})$
(C) $(A+B)(\overline{A}+C)(B+C)=(\overline{A}+\overline{B})(A+\overline{C})$
(D) $\overline{A}\,\overline{B}+A\,\overline{C}+\overline{B}\,\overline{C}=\overline{A}\overline{B}+A\,\overline{C}$

Open solution

Marks 2 10 questions

Q27. GATE ECE 2022

Consider a Boolean gate (D) where the output Y is related to the inputs A and B as, Y = A + $\overline{B}$, where + denotes logical OR operation. The Boolean inputs '0' and '1' are also available separately. Using instances of only D gates and inputs '0' and '1', _____ (select the correct options).

(A) NAND logic can be implemented
(B) OR logic cannot be implemented
(C) NOR logic can be implemented
(D) AND logic cannot be implemented

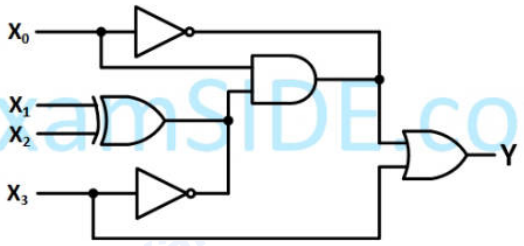
Open solution

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Q28. GATE ECE 2018

Open solution

The logic gates shown in the digital circuit below use strong pull-down nMOS transistors for LOW logic level at the outputs. When the pull-downs are off, high-value resistors set the output logic levels to HIGH (i.e. the pull-ups are weak). Note that some nodes are intentionally shorted to implement "wired logic". Such shorted nodes will be HIGH only if the outputs of all the gates whose outputs are shorted are HIGH.



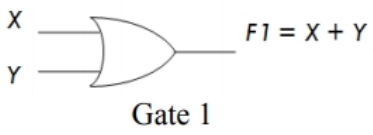
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The number of distinct values of $X_3X_2X_1X_0$ (out of the 16 possible values) that give $Y = 1$ is _____.

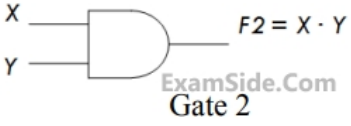
Q29. GATE ECE 2015 Set 3

Open solution

A universal logic gate can implement any Boolean function by connecting sufficient number of them appropriately. Three gates are shown.



Gate 1



ExamSide.Com

Gate 2



Gate 3 ExamSide.Com

Which one of the following statesments is TRUE?

(A) Gate 1 is a universal gate.

(B) Gate 2 is a universal gate.

(C) Gate 3 is a universal gate.

(D) None of the gates shown is a universal gate.

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Q30. GATE ECE 2015 Set 1

A 3-input majority gate is defined by the logic function M (a,b,c) = ab+bc+ca. Which one of the following gates is represented by the function M $\left(\overline{M(a,b,c)}, M(a,b,\overline{c}), c\right)$?

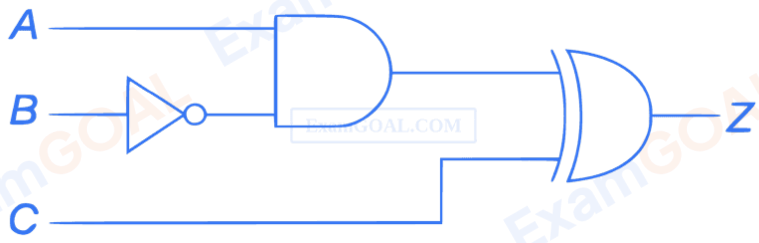
Open solution

(A) 3-Input NAND gate
(B) 3-Input XOR gate
(C) 3-Input NOR gate
(D) 3-Input XNOR gate

Q31. GATE ECE 2015 Set 1

All the logic gates shown in the figure have a propagation delay of 20 ns. Let A = C = 0 and B = 1 until time t = 0. At t = 0, all the inputs flip (i.e., A = C = 1 and B = 0) and remain in that state. For t > 0, output Z = 1 for a duration (in ns) of _____.

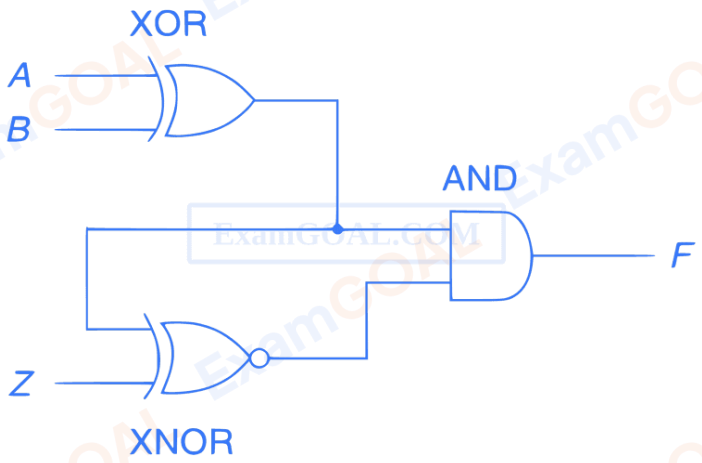
Open solution



Q32. GATE ECE 2014 Set 1

The output F in the digital logic circuit shown in the figure is

Open solution



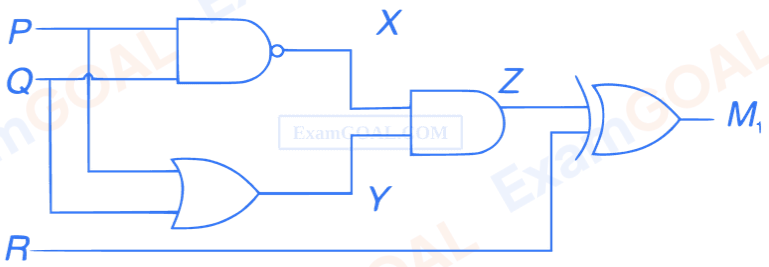
(A) $F = \overline{X}YZ + X\overline{Y}Z$
(B) $F = \overline{X}Y\overline{Z} + X\overline{Y}\,\overline{Z}$
(C) $F = \overline{XY}Z + XYZ$
(D) $F = \overline{XYZ} + XYZ$

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Q33. GATE ECE 2008

Which of the follwing Boolean expression correctly represents the relation between P, Q, R and M_1?

Open solution



(A) M_1=(P OR Q) XOR R

(B) M_1=(P AND Q) XOR R

(C) M_1=(P NOR Q) XOR R

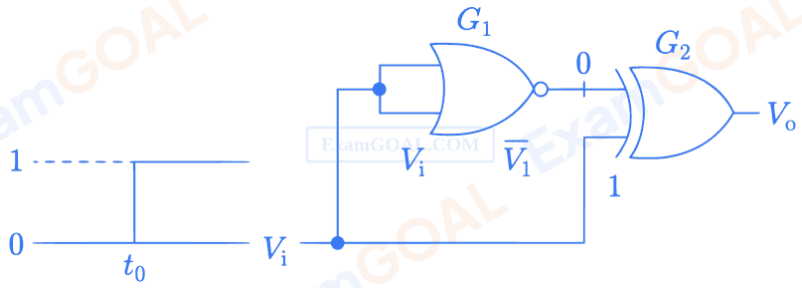
(D) M_1=(P XOR Q) XOR R

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Q34. GATE ECE 2002

The gates G_1 and G_2 in figure have propagation delays of 10nsec and 20nsec respectively. If the input V_i makes an abrupt change from logic 0 to 1 at time t = t_0, then the output waveform V_0 is

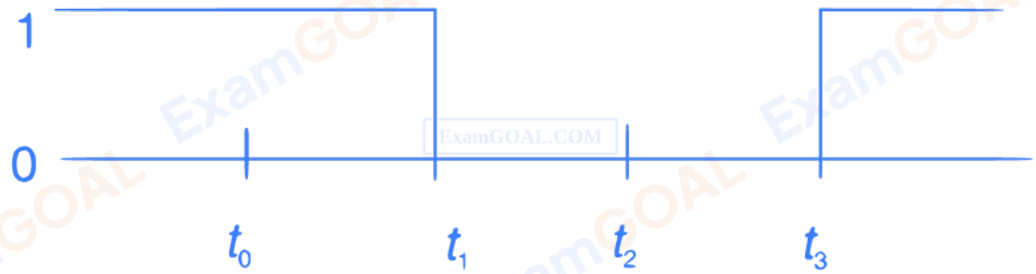
Open solution



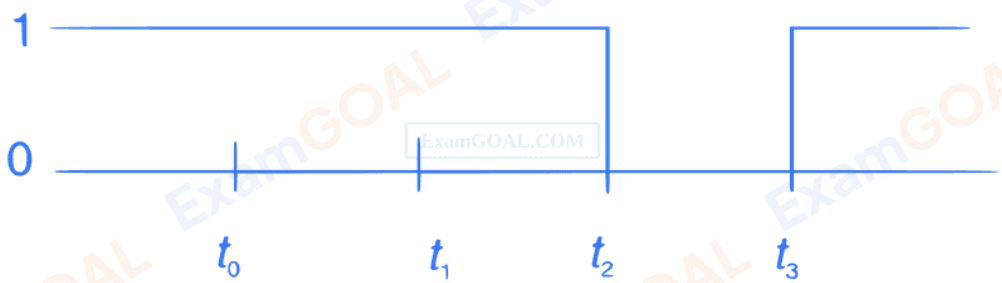
(A)



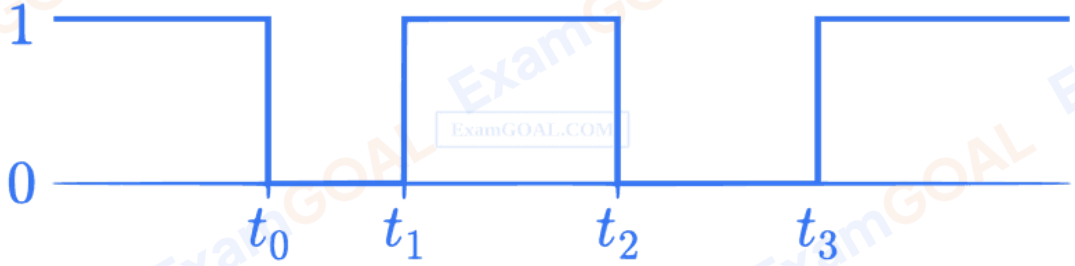
(B)



(C)



(D)

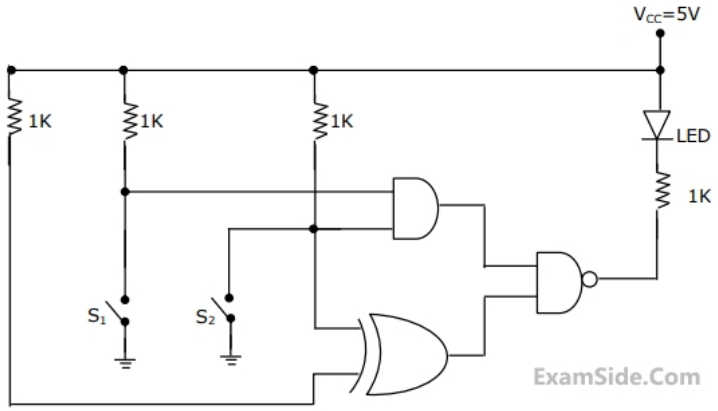


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Q35. GATE ECE 2001

Open solution

In the figure the LED



(A) emits light when both S1 and S2 are closed.

(B) emits light when both S1 and S2 are open.

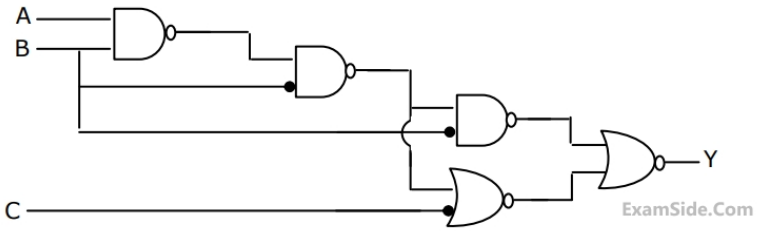
(C) emits light when only S1 or S2 is closed.

(D) does not emit light, irrespective of the switch positions.

Q36. GATE ECE 2000

Open solution

For the logic circuit shown in the figure, the simplified Boolean expression for the output Y is



(A) A + B + C (B) A (C) B (D) C

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Marks 5 3 questions

Q37. GATE ECE 2001

Open solution

For the digital block shown in Figure. 2(a), the output Y=f(S_3,S_2,S_1,S_0) where S_3 is MSB and S_0 is LSB. Y is given in terms of minterms as $Y = \sum m\,(1, 5, 6, 7, 11, 12, 13, 15)$ and its complements $\overline{Y} = \sum m\,(0, 2, 3, 4, 8, 9, 10, 14)$.

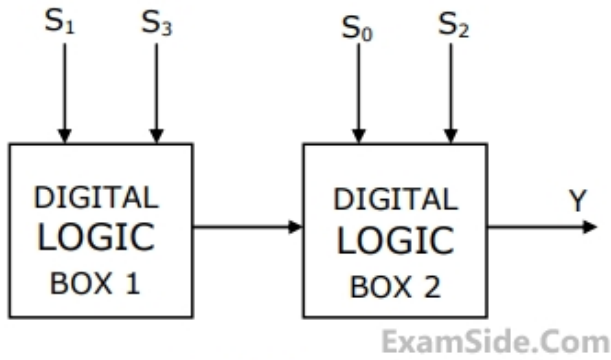


Figure 2(a)

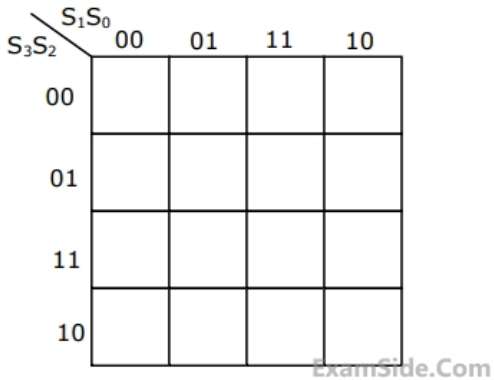


Figure 2(b)

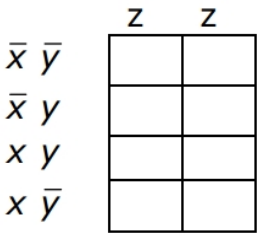
(a) Enter the logical values in the given Karnaugh map [figure2(b)] for the output Y.
(b) Write down the expression for Y in sum-of products from using minimum number of terms.
(c) Draw the circuit for the digital logic boxes using four 2-input NAND gates only for each of the boxes.

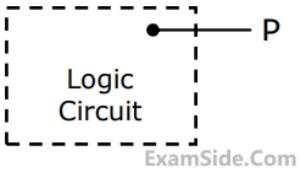
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Q38. GATE ECE 2000

The operating conditions (ON = 1, OFF = 0) of three pumps (x,y,z) are to be monitored. x = 1 implies that pump X is on. It is required that the indicator (LED) on the panel should glow when a majority of the pumps fail.

Open solution





(a) Enter the logical values in the K-map in the format shown in figure 3(a). Derive the minimal Boolean sum-of-products expression whose output is zero when a majority of the pumps fail.
(b) The above expression is implemented using logic gates, and point P is the output of this circuit, as shown in figure 3(b). P is at 0 V when a majority of the pumps fails and is at 5 V otherwise. Design a circuit to drive the LED using this output. The current through the LED should be 10 mA and the voltage drop across it is 1V. Assume that P can source or sink 10 mA and a 5 V supply is available.

Q39. GATE ECE 1993

The truth table for the output Y in terms of three inputs A, B and C are given in table. Draw a logic circuit realization using only NOR gates.

Open solution

A	0	1	0	1	0	1	0	1
B	0	0	1	1	0	0	1	1
C	0	0	0	0	1	1	1	1
Y	1	1	1	0	1	0	0	0

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Marks 8 1 questions

Q40. GATE ECE 1992

A combinational circuit has three inputs A, B and C and an output F. F is true only for the following input combinations?
A is false and B is true
A is false and C is true
A, B and C are all false
A, B and C are all true.

(a)Write the truth table for F. use the convention, true = 1 and false = 0.
(b)Write the simplified expression for F as a Sum of Products.
(c) Write the simplified expression for F as a product of Sums.
(d) Draw a logic circuit implementation of F using the minimum number of 2 input NAND gates only.

Open solution

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Answer Key

Marks 1

1. **A**	2. **A**	3. **2**	4. **C**	5. **A**
6. **C**	7. **A**	8. **A**	9. **C**	10. **B**
11. **D**	12. **D**	13. **B**	14. **C**	15. **A, D**
16. **C**	17. **A**	18. **100**	19. **B, C**	20. **B**
21. **B, D**	22. **B**	23. **B**	24. **B**	25. **C**
26. **A**				

Marks 2

27. **A, C**	28. **8 to 8**	29. **C**	30. **B**	31. **40**
32. **A**	33. **D**	34. **C**	35. **D**	36. **C**

Marks 5

37. **(b)** $Y = S_3S_2\overline{S_1} + \overline{S_3}S_2S_1 + S_3\overline{S_2}S_1 + S_3S_2S_0 + S_3S_1S_0$ **(c)** $Y = S_2\left[S_3 \oplus S_1\right] + S_0\left[\overline{S_3 \oplus S_1}\right]$	38. **Solve it**	39. $Y = \left(\overline{A} + \overline{C}\right)\left(\overline{A} + \overline{B}\right)\left(\overline{B} + \overline{C}\right)$

Marks 8

40. **(b) F =** $\overline{A} + BC$ **(c) F =** $\left(\overline{A} + B\right)(A + C)$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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